

Learn by looking and listening

A 45-minute session and optional night observation, adaptable for Peruvian classrooms and visiting groups.

Safe observation

Never look directly at the Sun or search for it with binoculars. For Inti, observe shadows or images.

1. Begin with what we know

10 min. Collect sky names in the group's languages. Introduce Inti, Killa, Collca, and Mayu. Ask who taught each name and where.

2. Set the place and time

15 min. In desktop Stellarium: F6, location; F5, date/time; F3, search. Try Moon, M45, and Alpha Centauri. On mobile or web, use the corresponding menus.

3. Observe, draw, compare

20 min. Draw the horizon and what you actually observe. Distinguish your observation from the account you learned. If cloudy, use a simulation and label it.

Prepare your outing

Allow 20-30 minutes for your eyes to adjust; dim your screen. Begin with the unaided eye. Arrange access and timing with local hosts or guides. Ask before recording stories or photographing people.

Names & counterparts in this collection

Not everything is a star: we distinguish planets, moons, clusters, galaxy, and dark dust figures. Names and accounts vary by community and period.

Inti | Sun

Star · smithsonian, gloria

Killa | Moon

Natural satellite · gloria

Chasca | Venus

Planet · garcilaso

Collca | Pleiades · M45

Star cluster · smithsonian, gloria, pucp

Llamaq ñawin | Alpha Centauri

Star system · pucp, gloria

Llamaq ñawin | Beta Centauri

Star system · pucp, gloria

Mayu | Milky Way

Galaxy / sky band · gloria

Llama | Dark-cloud llama

Dark constellation · gloria, pucp

Yutu | Tinamou / partridge

Dark constellation · gloria

Hanp'ata | Toad

Dark constellation · gloria

Machacuay | Snake

Dark constellation · gloria

About the names

Quechua is a family of varieties. We retain documented spellings: they are not a single orthography for every community. Collca does not have one identification across all sources.

Observation & listening sheet

Name / group:

Date and local time (Peru: UTC-5):

Place and direction you are facing:

Clouds, lights, and presence of the Moon:

Direct observation or simulation?:

Draw the horizon and what you actually see. Add a direction.

What did you observe? What did you learn from a story?

For a story: record community, language, source, and permission.

Sources & credits

smithsonian | Smithsonian · Reciprocity—Andean Style

National Museum of the American Indian: Inti, Colca, and Andean celebrations. Cultural context for living festivals.

[Open source](#)

gloria | GLORIA · Astronomy of the Incas, pp. 9–10

GLORIA project educational introduction (2014), pp. 9–10: examples of names and dark-cloud figures. A synthesis, not testimony from every community.

[Open source](#)

garcilaso | Garcilaso · Comentarios reales, Book III, ch. XXI

Comentarios reales, first part, book III, chapter XXI. Colonial source recording Chasca for Venus; requires critical historical reading.

[Open source](#)

pucp | PUCP · Anthropologica, note on Llamacñawin

Anthropologica (PUCP) article, with identifications and variants including Llamacñawin. Records more than one star group under Collca.

[Open source](#)

nasa | NASA · Stargazing for Beginners

Advice on naked-eye observation, dark adaptation, and preparation. Modern practical astronomy.

[Open source](#)

stellarium | Stellarium · User Guide

Official planetarium guide: location, date, time, and object search. A simulation should be identified as such.

[Open source](#)

minedu | MINEDU · Urin qichwa qillqay

MINEDU, Urin qichwa qillqay: reference on Southern Quechua writing. This does not mean the ministry reviewed or endorsed Mayu's translations.

[Open source](#)

Language

The Quechua version uses a proposed Cusco-Collao rendering and needs linguistic review. Entries retain source spellings; we do not certify pronunciation or community translation.